

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

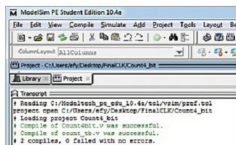


Fig. 6: Compilation window

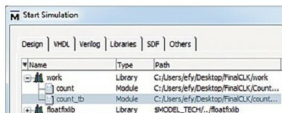


Fig. 7: Library tab

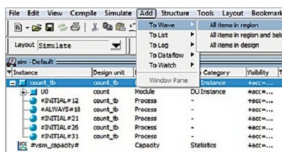


Fig. 8: Add wave to the project

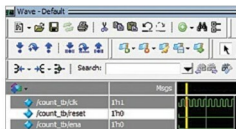


Fig. 9: Wave window

The source code (Count4bit.v) of 4-bit counter and its test bench that consists of a clock generator is explained next.

To start 4-bit counter-clock-generator simulation, install ModelSim V10.4a on a Windows PC and follow

the steps mentioned below.

1. Start ModelSim from the desktop; you will see ModelSim 10.4a dialogue window.

2. Create a project by clicking Jump Start on Welcome screen.

3. Create Project window pops up

(Fig. 1). Select a suitable name for your project. Set project location and leave rest as default, followed by clicking OK.

4. Add items to the Project window pops up (Fig. 2).

5. On this window, select Create New File option.

6. Give an appropriate file name (say, Count4bit.v) for the file you want to add, and choose Verilog under Add file as type and Top Level under Folder (Fig. 3).

7. On the workspace section of the main window (Fig. 4), double-click on the file you have just created (Count4bit.v in our case).

8. Type your Verilog code in the new window. This is the code for 4-bit counter. Our main goal is to write a self-test bench that will generate clock automatically for the simulation output window. Save your code from File menu.

Now add new file to Count4bit.v by right-clicking. Select Add to Project→New File as shown in Fig. 5.

Give file name count_tb.v. This is the test case which consists of clock generator, reset and enable controls. Follow the steps from 7 through 8 as mentioned above.

Details of this clock generator of the 4-bit counter can be understood from the relevant source files.

Compiling/debugging project files.

1. Select Compile→Compile All.

2. Compilation result is shown on the main window. A green tick is shown against each file name, which means there are no errors in the project (Fig. 6).

Simulating clock generator for 4-bit counter.

1. Select Start Simulate→ Start Simulation. Click on Library menu from the main window and then click on plus (+) sign next to the work library. You should see the name that we have just compiled in Fig. 6.

2. In work library, select tb_count and click on OK (Fig. 7). This will open sim-default window as shown in Fig. 8.

3. Go to Add→To Wave→ All items in region.

4. Monitor the clock generator signal for simulation purposes.

5. The test bench is used to verify clock generation for counter design. We are now ready to simulate our design by clicking Run from Simulate menu bar as shown in Fig. 8.

You can view the output on the Wave window as shown in Fig. 9. **EFY**



Arnab Bansal is an electronics hobbyist and loves to tinker with circuit designs

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